

Project Proposal

FreqEasy-ML, Machine Learning-Based Acoustic Room Assessment Assistant

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1 BACKGROUND

1.1 PROBLEM STATEMENT

Small recording, learning, rehearsal, and production spaces often suffer from poor acoustic performance due to untreated reflections, standing waves, low-frequency buildup, background noise, and changing environmental conditions. These issues can reduce speech clarity, recording quality, and the overall usability of the space. Although acoustic measurements such as frequency data can provide useful information about room performance, these values are difficult for non-expert users to interpret without prior knowledge of acoustics.

As a result, users may not know whether a room requires wall panels, bass traps, ceiling clouds, reflection-point treatment, or other forms of acoustic treatment. They may also struggle to understand how the intended use of the room, such as vocals, drums, guitar, podcasting, classroom instruction, rehearsal, or general music production, affects the type of acoustic treatment required.

This project addresses the gap between raw acoustic measurement data and practical treatment decisions by applying supervised machine learning to acoustic room assessment. The proposed system will analyze Fast Fourier Transform (FFT) or frequency profiles to make conclusions about the source and its surrounding area. This dataset will be used to train and evaluate machine learning models that classify common acoustic conditions and translate the classification results into practical treatment recommendations.

The goal of the project is to make acoustic analysis more accessible by using measured acoustic features and room characteristics to predict acoustic issue categories such as low-frequency buildup, high background noise, reflection-related issues, or acceptable acoustic performance. These outputs can then support user-friendly recommendations such as bass traps, wall panels, ceiling clouds, or reflection-point treatment. This reduces the need for expert acoustic knowledge while supporting better recording, learning, and listening environments.

1.2 PROJECT DESCRIPTION

FreqEasy-ML is a machine learning-based acoustic room assessment system designed to help non-expert users evaluate and improve small recording, learning, rehearsal, and production spaces. The system builds on the Smart Distributed Acoustic Calibration System (SDACS) which uses distributed sensor nodes to collect acoustic and environmental data from different room locations. Each node will measure FFT frequency peaks from 20Hz to 20kHz and represent them with gains in decibels (dB).

The main AI/ML task will be supervised classification. Using labelled measured (and synthetic) FFT data, the system will train and evaluate models such as decision trees, random forests, support vector machines, or similar classifiers [5]. The model will predict acoustic issue categories such as low-frequency buildup, excessive background noise, reflection-related issues, or acceptable acoustic performance. These classifications will then be translated into treatment recommendations such as bass traps, wall panels, ceiling clouds, reflection-point treatment, or general acoustic improvements.

A rule-based method may be used as a baseline for comparison, but the primary deliverable will be the supervised ML classifier and its evaluation. This makes the project a true AI/ML application because it uses labelled examples to learn relationships between room conditions, acoustic features, and treatment recommendations instead of relying only on fixed threshold rules.

The initial Machine Learning (ML) model will run on a computer, Raspberry Pi, Node-RED backend, or Flutter-side processing environment. This keeps the project achievable while still demonstrating machine learning. TinyML deployment on the ESP32-S3 for SDACS may be explored as future work after the dataset, model performance, and system benchmarks have been validated [6].

The novelty of FreqEasy-ML is that it combines complex frequency profile data and treatment recommendation logic into one accessible acoustic support tool. The system provides a guided workflow from measurement to classification to recommendation, making it a suitable ML project that integrates complex acoustic analysis and simple, user-friendly recommendations.

2 LITERATURE REVIEW

2.1 EXISTING SOLUTIONS COMPARATIVE STUDY

Several existing solutions are available for evaluating room acoustics, but most either provide raw measurements without interpretation or require a high level of acoustic knowledge.

Existing Solution	Function	Limitation	How FreqEasy-ML Improves on It
Smartphone sound apps	Provide low-cost access to basic sound level or frequency measurements.	Accuracy depends on the phone microphone and calibration; interpretation is still left to the user.	Uses a trained model that can filter out or notify the user of poor data quality.
Room EQ Wizard and similar acoustic software	Provides detailed frequency response and acoustic analysis tools.	Powerful but technical; users must understand measurement setup, frequency response, and treatment decisions.	Converts acoustic features into predicted issue categories and user-friendly recommendations.
Professional acoustic consultants	Provide expert analysis and treatment recommendations.	Can be expensive and less accessible for students, hobbyists, small studios, or classrooms.	Provides low-cost first-pass acoustic guidance using measured data and ML classification.
Commercial room correction systems	Adjust playback response using EQ or calibration microphones.	Often focus on electronic correction rather than physical acoustic treatment.	Recommends physical treatment categories such as wall panels, bass traps, or ceiling clouds.

Compared with these solutions, FreqEasy-ML is intended to provide an accessible bridge between acoustic measurement and practical decision-making. The system will not replace a professional acoustic consultant, but it can help non-expert users understand common acoustic issues in frequency amplification and select appropriate first-stage treatment options using measured features and supervised machine learning classification.

2.2 RELEVANT STUDIES

2.2.1 *Living Walls and Acoustics [1]*

In a 2022 study published in the *Journal of Building Engineering*, researchers F. Scamoni, C. Scrosati, M. Depalma, and B. Barozzi investigated the acoustic performance and long-term durability of a novel indoor living wall system. The team conducted experimental evaluations in a controlled laboratory environment to address the growing need for sustainable, health-promoting architectural features that can simultaneously mitigate indoor noise pollution. By analyzing the system's sound absorption and airborne sound insulation properties over an extended period, the authors sought to determine if integrating vegetation into indoor building elements could serve as a viable, long-lasting solution for improving room acoustics while maintaining structural and botanical integrity.

2.2.2 *Smartphone-Based Space Mapping [2]*

In June 2018, researchers S. Pradhan, G. Baig, W. Mao, L. Qiu, G. Chen, and B. Yang developed and evaluated "BatMapper," a system designed for smartphone-based acoustic indoor space mapping. Published in the *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, the study aimed to overcome the limitations of traditional, expensive specialized hardware used for indoor floor plan generation. By utilizing the built-in microphones and speakers of everyday smartphones to emit acoustic signals and measure their echoes, the authors successfully created a low-cost, accessible, and autonomous method for ordinary users to map indoor structural boundaries, thereby enhancing navigation and location-based services in indoor environments.

2.2.3 *University Space Acoustic Mapping [3]*

In a 2024 study published in *E3S Web of Conferences*, S. Jha, A. Gupta, A. Gupta, R. K. Mishra, and R. P. S. Jadon analyzed and mapped the indoor acoustic environments of university study spaces. The researchers conducted their fieldwork within a university campus to address the critical issue of how ambient noise levels impact student concentration, cognitive performance, and overall well-being. By measuring and mapping noise variations across different educational zones, the authors provided data-driven insights aimed at helping academic institutions optimize the design and management of their learning spaces for better academic productivity.

2.2.4 Autonomous Microphone Localization using VoiceMap [4]

In January 2024, researchers Sheng Chen, Renrui Tan, Zhiming Wang, Xinyu Tong, and Keqiu Li introduced "VoiceMap," an autonomous system for mapping microphone arrays to improve voice localization. Published in the IEEE Internet of Things Journal, the study addresses the complex and tedious problem of manually calibrating and locating the positions of distributed microphones in smart indoor environments. To solve this, the authors designed a framework that utilizes ambient human voice and sound cues to automatically self-localize and map the microphone positions, significantly improving the efficiency and accuracy of voice-controlled IoT applications and smart home technologies.

2.2.5 Machine Learning for Acoustic Classification [5]

Salamon and Bello demonstrated that machine learning can be applied to environmental sound classification using labelled audio data and extracted acoustic features [5]. Their work is relevant to FreqEasy-ML because the proposed system will also use measured acoustic features to classify meaningful acoustic conditions. Although FreqEasy-ML is focused on room acoustic issues rather than urban sound events, the same general idea applies: acoustic data can be processed into features and used to train a supervised classifier. This supports the project's use of features such as RMS level, dB SPL, dBFS, FFT frequency peaks, frequency band energy, and noise-floor-related measurements as inputs to a machine learning model.

2.2.6 TinyML and Edge AI for Acoustic Analysis [6], [7]

TinyML and edge AI provide a possible future path for running trained machine learning models directly on embedded devices. Google's LiteRT for Microcontrollers documentation describes a runtime designed for microcontrollers and other memory-constrained devices [6]. Edge Impulse also provides tools for building, training, and deploying machine learning models to edge devices [7]. These sources support the future-work direction of FreqEasy-ML, where a trained acoustic classifier could eventually be optimized and deployed directly on an ESP32-S3 sensor node. However, because TinyML requires careful consideration of memory, processing power, model size, and dataset quality, the current capstone will first focus on training and evaluating the classifier on a computer, Raspberry Pi, backend system, or application-side environment.

2.3 RATIONALE

FreqEasy-ML combines several major areas of electronic systems engineering and machine learning into one practical and testable system. The project requires digital signal processing, data collection, dataset creation, supervised machine learning, and real-world validation.

The project builds on the SDACS platform by using the frequency profile at a possible node, and analyzing the data for trends, issues, and noise as an additional service to the user. These nodes can process the FFT frequency peaks, frequency band energy, and noise-floor-related data. This makes the project more than a simple measurement tool because it connects real sensor data to the acoustic behaviour of an actual space.

The project is also valuable because it addresses a real-world problem. Small recording, rehearsal, classroom, and production spaces often have acoustic issues that are difficult for non-expert users to diagnose. Many users can measure sound levels or view frequency data but may not understand how those values relate to practical acoustic treatment decisions. FreqEasy-ML helps solve this problem by using supervised machine learning to classify acoustic conditions and translate the results into clear and low budget-friendly recommendations.

A key engineering challenge is the communication and data pipeline. The acoustic trends must be reliable, requiring a large diverse dataset that is labelled accurately for the algorithm to learn effectively. Additionally, the results must be communicated in a clear way for a broad user narrative, a client with little-to-no acoustic knowledge.

The machine learning component provides a clear AI/ML deliverable. The project will create a labelled dataset from SDACS measurements and synthetic datasets, train candidate supervised learning models, evaluate model performance, and compare the classifier against a rule-based baseline.

Overall, FreqEasy-ML has strong merit as a capstone project because it integrates embedded systems, acoustic sensing, MQTT networking, data analysis, supervised machine learning classification, recommendation logic, and real-world validation. The system can be tested in actual rooms by comparing measured acoustic data, assigned labels, model predictions, and expected room behaviour.

3 SYSTEM OVERVIEW

3.1 PROJECT DEFINITION

FreqEasy-ML is a supervised machine learning system that uses FFT data / graphs to classify common acoustic conditions in small rooms. The system will use FFT peak frequency, gain (dB), and noise-floor-related measurements.

The main machine learning task is classification. The system will train and evaluate models that classify a room measurement into acoustic issue categories such as low-frequency buildup, high background noise, reflection-related issues, or acceptable acoustic performance. The classification result will then be translated into a practical recommendation category such as bass traps, wall panels, ceiling clouds, reflection-point treatment, or general acoustic improvement.

The scope of the project includes acoustic FFT data collection, dataset creation, model training, model evaluation, and display of the classification result through the user interface. The initial ML model will run on a computer, Raspberry Pi, Node-RED backend, or Flutter-side processing environment. TinyML deployment on the sensor node (microcontroller) will be considered future work after the dataset, classifier performance, and system benchmarks have been validated.

Success criteria for the project include:

- i. Collecting acoustic and environmental data from at least one SDACS node
- ii. Extracting useful acoustic features such as SPL, RMS, dBFS, FFT peaks, and frequency band information
- iii. Creating a labelled dataset using measured acoustic data and room/context information
- iv. Training and testing at least two supervised ML models
- v. Comparing ML model performance against a rule-based baseline
- vi. Displaying the predicted acoustic issue category and treatment recommendation through the user interface

3.2 SYSTEM BLOCK DIAGRAM

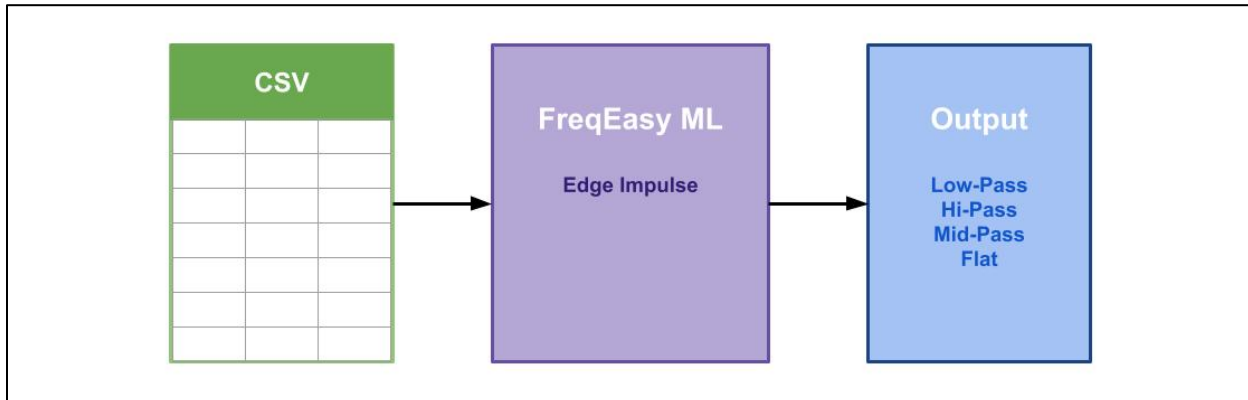


Figure 1: FreqEasy System Block Diagram

The proposed FreqEasy ML will take a CSV with FFT feature data, process it using Edge Impulse, and produce one of four features:

- i. Noise Level: Noisy or Quiet
- ii. Frequency Profile: Low-Pass, Mid-Pass, Hi-Pass
- iii. Speech Present: Yes or No

3.3 EDGE IMPULSE INTEGRATION

Edge Impulse will be used as the primary machine learning development platform for FreqEasy-ML. Edge Impulse is a cloud-based TinyML and edge AI development environment that supports data collection, dataset management, feature extraction, model training, testing, and deployment. The platform simplifies the machine learning workflow by providing a graphical interface for building and evaluating classification models without requiring extensive machine learning programming experience.

For this project, FFT-derived acoustic features generated by the SDACS system will be stored in comma-separated value (CSV) files. Each dataset entry will contain extracted acoustic features such as sound pressure level (SPL) and other relevant measurements. These labelled datasets will be uploaded into Edge Impulse as tabular data for supervised learning.

The Edge Impulse workflow will consist of four primary stages:

1. **Dataset Creation** – Collection of measured and synthetic FFT-based acoustic data representing different room acoustic conditions.

2. **Feature Processing** – Importing FFT-derived features into Edge Impulse and preparing them for machine learning classification.
3. **Model Training and Evaluation** – Training supervised classification models and evaluating their performance using metrics such as accuracy, validation loss, and confusion matrices.
4. **Model Deployment** – Exporting the trained model as an Arduino or C++ library for future deployment on the ESP32-S3 microcontroller or integration within the Raspberry Pi backend.

Initially, the machine learning model will run on a computer, Raspberry Pi 5, or backend application while the system is being developed and tested. This will allow the team to evaluate the model's accuracy and ensure it can correctly classify acoustic conditions and generate useful recommendations. Once the dataset and model have been validated, a future version of the project may use Edge Impulse to deploy the model directly onto the ESP32-S3 sensor nodes. Running the model on the microcontroller would allow acoustic classifications to be performed locally, reducing network traffic and providing faster results.

3.4 FUTURE WORK AND LIMITATIONS

The main limitation of FreqEasy-ML is the need for a reliable labelled acoustic dataset. The classifier will only be as accurate as the data used to train and test it. Collecting data from many room types, dimensions, treatment conditions, and use cases would improve model generalization, but this may be limited within the capstone timeline.

Another limitation is that acoustic treatment recommendations may depend on factors that are difficult to fully capture using basic FFT features alone. For example, room geometry, wall materials, furniture placement, and exact microphone/speaker positioning can influence acoustic results. For this reason, the system should present recommendations as guidance rather than certified acoustic design.

TinyML deployment is a future enhancement rather than the main project deliverable. A future version of the system could compress or optimize the trained model and deploy it directly on the ESP32-S3 sensor nodes [6]. This would allow local inference, reduce communication requirements, and improve response time. However, TinyML requires careful consideration of

RAM, flash memory, CPU performance, model size, and power consumption. The current capstone will first focus on collecting data, training and evaluating a supervised ML model, and validating the recommendation workflow before attempting on-node TinyML inference.

4 IMPACTS ON SOCIETY AND PROFESSIONAL RESPONSIBILITY

FreqEasy-ML has potential societal, educational, and economic value because it makes acoustic analysis more accessible to non-expert users. Students, musicians, podcasters, teachers, and small studio owners may not have the budget or technical knowledge to hire an acoustic consultant or use advanced acoustic analysis software. By translating measured acoustic data into practical classification results and treatment recommendations, the system can support better recording, learning, communication, and listening environments.

The project also has industrial and entrepreneurial value because it applies complex data analysis and machine learning to a real-world problem. A low-cost acoustic assessment system could be useful in small studios, classrooms, rehearsal rooms, offices, and home production spaces.

Professional responsibility is important in this project. The system should avoid presenting recommendations as guaranteed or certified acoustic designs. Instead, the output should be framed as decision-support guidance based on measured data and model predictions. Privacy should also be considered because acoustic systems may involve microphones. Where possible, the system should store extracted acoustic features rather than raw audio recordings. Environmental responsibility should also be considered by recommending practical treatment options and avoiding unnecessary material use.

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